subnational postal zone if the ZIP code is not available.[264] One commenter recommended using geographic coordinates to describe the location of assets subject to a material physical risk because they would better fit climate models.[265] Another commenter recommended requiring the disclosure of specific addresses, and not just ZIP codes, to identify the location of assets subject to a material physical risk to enable investors to fully assess the registrant's exposure to the physical risk.[266] This commenter also urged the Commission to require the proposed disclosure with respect to all of a registrant's locations that are material to its businesses rather than only the locations subject to a physical climate risk, stating that physical climate risk potentially impacts a registrant at all of its locations.[267]

Several commenters supported the proposed requirement to disclose the percentage of assets that are located in flood hazard areas if a registrant has determined that flooding is a material physical risk.[268] Several commenters also supported the proposed requirement to disclose the amount of assets (e.g., book value and as a percentage of total assets) located in regions of high or extremely high water stress, and the percentage of the registrant's total water usage from water withdrawn in those regions, if a registrant has determined that high or extremely high water stress is a material physical risk.[269] Commenters stated that the proposed disclosure requirements would help investors understand the extent of the water-related risk to

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264 *See, e.g.*, letters from Amer. for Fin. Reform, Evergreen Action *et al*.; IATP; and TotalEnergies.

265 *See* letter from CalSTRS.

266 *See* letter from Wellington Mgmt.

267 *See id.*

268 *See, e.g.,* letters from Anthesis Group (June 16, 2022) ("Anthesis"); CalPERS; Domini Impact; Eni SpA; ERM CVS (June 17, 2022); IAA; Moody's; Morningstar; NRDC; PRI; TotalEnergies; and Wellington Mgmt.

269 *See, e.g.,* letters from Anthesis; CalSTRS; Domini Impact; ERM CVS; IAA; Moody's; Morningstar; Paradice Investment Management (June 17, 2022) ("Paradice Invest. Mgmt."); TotalEnergies; and Wellington Mgmt.

which a registrant is exposed.[270] Some commenters generally stressed the importance to investors of obtaining quantitative data from registrants about the physical risks to which they are subject and recommended that the Commission require registrants to similarly provide the percentage of assets or other quantitative data relevant to assessing a registrant's exposure to other material physical risks, such as heatwaves, droughts, and wildfires.[271]

With regard to flooding risk disclosure, some commenters recommended that the Commission require the use of Federal Emergency Management Agency's ("FEMA's") flood hazard terminology and maps to help further the comparability of the disclosure.[272] One commenter recommended the use of a different flood model that it believed was more up-to-date and more comprehensive than FEMA's flood mapping.[273] Another commenter supported an approach that would allow for different definitions of "flood hazard area" or "water-stressed area" to be used as long as the registrant disclosed the source of the definitions together with the methodologies and assumptions used in disclosing the water-based physical risk.[274]

Several commenters supported the proposed provision requiring a registrant to describe the nature of an identified transition risk, including whether it relates to regulatory, technological, market (including changing consumer, business counterparty, and investor preferences), liability, reputational, or other transition-related factors, and how those factors impact the registrant.[275] Some commenters also supported the proposed definition of transition

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270 *See, e.g.*, letters from ERM CVS; IAA; Moody's; and Morningstar.

271 *See, e.g.*, letters from Anthesis; CalPERS; IAA; and Morningstar.

272 *See, e.g.*, letters from Anthesis; NRDC; and PRI.

273 *See* letter from CalPERS (recommending use of the First Street Foundation Flood Model).

274 *See* letter from Moody's; *see also* letter from Wellington Mgmt. (stating that, if address-specific locations are not required, the Commission should require the disclosure of methodologies and data sources used for flooding disclosure).

275 *See, e.g.*, letters from Eni SpA; Moody's; Morningstar; SKY Harbor; TotalEnergies; and Wellington Mgmt.

risk.[276] Several commenters stated that the Commission should include additional examples within the definition of transition risk, including the risk of impacts on local and indigenous communities and workers caused by a transition to a lower carbon economy.[277]

Several commenters supported including the negative impacts on a registrant's value chain in the definition of climate-related risk, as proposed.[278] One commenter stated that because information concerning climate-related risks involving a registrant's value chain may be more important to investors than such risks involving a registrant's own operations, disclosure of climate-related risks in the value chain should be an integrated part of the broader disclosures about the material climate-related risks management is assessing, managing, and reporting to the board, despite the difficulty of providing such value chain information.[279] Another commenter stated that it supported including value chain impacts in the definition of climate-related risk as long as such impacts relate to direct impacts on a registrant's operations.[280] Some commenters also supported the proposed definition of value chain to mean the upstream and downstream

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276 *See, e.g.*, letters from Eni SpA; Morningstar; SKY Harbor; and TotalEnergies.

277 *See, e.g.*, letters from Boston Common Asset Mgmt.; CalPERS; Domini Impact; IAA; and ICCR.

278 *See, e.g.*, letters from Amer. for Fin. Reform, Sunrise Project *et al*.; Anthesis; Domini Impact; D. Hileman Consulting; Eni SpA; Morningstar; NY SIF; PRI; PwC; TotalEnergies; US Technical Advisory Group to TC207 (June 17, 2022) ("US TAG TC207"); and Wellington Mgmt.

279 *See* letter from PwC. This commenter provided the following examples of when climate-related risks involving a registrant's value chain may be more important to investors than such risks involving the registrant's own operations: the manufacturer of "a product reliant on a rare mineral for which mining may be limited due to emissions created in extraction, precursor manufacturing, and transport, or, alternatively, a lender whose primary business is financing emissions-intensive operations."

280 *See* letter from Eni SpA.

activities related to a registrant's operations.[281] One commenter stated that the definition of value chain should be consistent with the definition provided by the GHG Protocol.[282]

Many other commenters opposed the proposed climate-related risk disclosure requirement.[283] Some of these commenters contended that the Commission's rules already require a registrant to disclose material climate risks, and that therefore there is no need for the proposed climate-related risk disclosure requirement.[284] Several other commenters stated that the proposed climate-related risk disclosure requirement would inundate investors with an extensive amount of granular information that is largely immaterial.[285] Commenters provided as an example of such immaterial disclosure the proposed requirement to disclose the ZIP codes of assets located in flood hazard areas or other regions in which a registrant's assets are subject to a material climate-related risk.[286] Some commenters stated that the highly detailed disclosure required by the proposed climate risk disclosure rule would confuse investors by causing them to believe that a climate-related risk is more important than other disclosed risks that are presented

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281 *See, e.g.*, letters from Amer. for Fin. Reform, Sunrise Project *et al.*; Anthesis; and Morningstar.

282 *See* letter from Morningstar; *see also* letter from D. Hileman Consulting (stating that if the Commission defines value chain, it should adopt a definition that is already well-established, such as the GHG Protocol's definition of value chain).

283 *See, e.g.*, letters from ABA; American Chemistry Council (June 17, 2022) ("Amer. Chem."); American Fuel and Petrochemical Manufacturers (June 17, 2022) ("AFPM"); Biotechnology Innovation Organization (June 17, 2022) ("BIO"); Business Roundtable; Chamber; Davis Polk (June 9, 2022); Fenwick West; GPA Midstream Association (June 17, 2022) ("GPA Midstream"); Insurance Coalition (June 17, 2022) ("IC"); Nareit (June 17, 2022) ("Nareit"); National Mining Association (June 17, 2022) ("NMA"); Retail Industry Leaders Association (June 17, 2022) ("RILA"); and Soc. Corp. Gov.

284 *See, e.g.*, letters from AFPM; BIO; and GPA Midstream.

285 *See, e.g.*, letters from ABA; Amer. Chem.; AFPM; Business Roundtable; Chamber; Davis Polk; Fenwick West; Nareit; NMA; RILA; SIFMA; and Soc. Corp. Gov.

286 *See, e.g.*, letters from ABA; Allstate Corporation (June 17, 2022) ("Allstate") ("Requiring information at a granular level such as ZIP code would create an operational burden and would produce an excessive amount of information that we expect would not be decision-useful for most investors."); Amer. Chem.; AFPM; BOA; Business Roundtable; Chamber; Davis Polk; NAM; Nareit; PGIM (June 17, 2022); RILA; SIFMA; and Soc. Corp. Gov.

in less detail.[287] Some commenters also stated that the overly granular disclosure elicited by the proposed rule would potentially require registrants to disclose competitively sensitive information.[288] Other commenters stated that, due to uncertainties in climate science, and uncertainties regarding some of the underlying concepts upon which the proposed climate risk disclosure requirement is based, the disclosure of material climate-related risks would be unduly burdensome for many registrants.[289] Another commenter stated that a registrant should only be required to disclose a climate-related risk that management is assessing, managing, and reporting to the board, rather than disclosing information regarding any climate risk.[290]

Several commenters also opposed the proposed disclosure requirements concerning the percentage of assets located in flood zones and similar quantitative data for assets located in high water-stressed areas.[291] One commenter stated that flood risks and high water-stress risks are not comparable within a firm, across sectors, and across regions of the country, so investors are unlikely to make investment decisions based on this information.[292] This commenter further stated that the Commission has not justified singling out risks relating to flooding and high water stress for detailed prescriptive disclosures, which dilutes the importance of other material

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287 *See, e.g.*, letters from ABA; Fenwick West; GPA Midstream; and Nareit.

288 *See, e.g.*, letters from IC; NAM; National Grid; RILA; and Soc. Corp. Gov.

289 *See, e.g.*, letters from NMA; and RILA; *see also* letter from IC (stating that the proposed climate risk disclosure requirement raises concerns for insurers because there is no consensus scientific method for insurers to distinguish between weather-related risks and climate-related risks).

290 *See* letter from PwC.

291 *See, e.g.*, letters from ABA; AFPM; BOA; and D. Hileman Consulting.

292 *See* letter from AFPM; *see also* letter from BOA (stating that investors would not be able meaningfully to compare water-stress risks across different companies without standard definitions for "high water-stress" and "extreme high water-stress.").

information.[293] One other commenter stated that the proposed flood risk requirement is not necessary because the majority of companies are not subject to such physical risk.[294] Other commenters stated that such granular disclosure for water-related physical risks would impose a heavy reporting burden for registrants and could raise competitive and security risk concerns.[295]

Several commenters also opposed the proposed transition risk disclosure requirement, including the proposed definition of transition risk.[296] Some commenters stated that the proposed requirement would result in overly granular disclosure that would not be decision-useful for investors and would be burdensome for registrants to produce.[297] One commenter stated that the proposed definition was overly broad and would require a registrant to make the difficult determination of whether a particular activity was undertaken to address a transition risk or was part of a registrant's normal business strategy.[298] Another commenter stated that it would be challenging for companies doing business in multiple markets to provide comparable, consistent, and reliable disclosure about transition risks given complex, dynamic, and varied global factors.[299] Other commenters stated that because the proposed definition of transition risk

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293 *See* letter from AFPM; *see also* letter from ABA (stating that by proposing highly prescriptive disclosure requirements, such as those based on flood hazard areas or assets of "high or extremely high water stress," the Commission may potentially narrow disclosures related to the full range of environmental or climate issues that are materially relevant to a registrant's business and strategy); and D. Hileman Consulting (stating that it is not necessary for the Commission to enumerate specific climate-related risks, such as flooding or water stress, as there is the risk that registrants could downplay other types of risk).

294 *See* letter from BIO.

295 *See, e.g.*, letters from CEMEX; and NAM.

296 *See, e.g.*, letters from Airlines for America (June 17, 2022); Allstate; Alphabet *et al*.; American Council for Capital Formation (June 17, 2022) ("ACCF"); Chamber; Enbridge Inc. (June 16, 2022) ("Enbridge"); Interstate Natural Gas Association of America (June 17, 2022) ("INGAA"); PwC; and United States Council for International Business (June 17, 2022) ("USCIB").

297 *See, e.g.*, letters from ACCF; and Allstate.

298 *See* letter from Alphabet *et al*.

299 *See* letter from USCIB.

would require a registrant to consider impacts on its value chain, the resulting disclosures are likely to be overly detailed and could obscure more important information.[300] One other commenter stated that the proposed transition risk disclosure requirement would be difficult to comply with because of the speculative nature of certain transition risks.[301] A different commenter stated that because of the broad definition of transition risk, the Commission should provide additional guidance regarding the scope of the transition risk disclosure requirement.[302]

Many commenters opposed including the negative impacts on a registrant's value chain in the definition of, and related disclosure requirement concerning, its climate-related risks.[303] Commenters stated that the proposed definition would impose impractical burdens on registrants by forcing them to obtain and assess climate risk information about their third-party suppliers and customers over which they have little to no control.[304] Commenters in the agricultural sector were particularly opposed to the proposed definition because it would impose costs and burdens on farmer and rancher suppliers, many of whom are private entities, to produce the information needed by registrants to comply with the proposed climate-related risk requirement.[305] Other commenters stated that, due to the inability to obtain such third-party information, the proposed disclosure requirement is likely to elicit boilerplate disclosure about the climate-related risks of a

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300 *See* letters from Airlines for America; and Chamber.

301 *See* letter from INGAA.

302 *See* letter from PwC.

303 *See, e.g.*, letters from Airlines for America; Arizona Farm Bureau Federation (June 17, 2022) ("AZ Farm"); California Farm Bureau (June 17, 2022) ("CA Farm"); Chamber; CEMEX;D. Burton, Heritage Fdn.; Energy Transfer LP (June 17, 2022) ("Energy Transfer"); Georgia Farm Bureau ("June 17, 2022) ("GA Farm"); GPA Midstream; HP; Indiana Farm Bureau (June 17, 2022) ("IN Farm"); National Agricultural Association (June 17, 2022) ("NAA"); Pennsylvania Farm Bureau (June 17, 2022) ("PA Farm); Soc. Corp. Gov.; United Airlines Holdings, Inc. (June 17, 2022) ("United Air"); Western Midstream; and Williams Cos.

304 *See, e.g.*, letters from CEMEX; GPA Midstream; HP; Soc. Corp. Gov.; United Air; Western Midstream; and Williams Cos.

305 *See, e.g.,* letters from AZ Farm; CA Farm; GA Farm; IN Farm; NAA; and PA Farm.

registrant's value chain.[306] Because of these concerns, several commenters requested that the Commission remove the concept of value chain from the scope of the climate risk disclosure requirement.[307] More generally, several commenters stated that any Commission climate risk disclosure requirement should be more principles-based and grounded on traditional notions of materiality.[308]

Many commenters supported the proposed definition of climate-related opportunities because it is consistent with the TCFD definition.[309] Many commenters also supported keeping the disclosure of climate-related opportunities optional, as proposed.[310] Some of these commenters expressed the view that, while disclosure of climate-related opportunities can provide insight into a registrant's management of climate-related risks and its related strategy, mandatory disclosure of climate-related opportunities could lead to greenwashing.[311] Some commenters, however, stated that disclosure of climate-related opportunities should be mandatory because such opportunities are frequently related to the reduction of climate-related

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306 *See, e.g.*, letters from Energy Transfer; HP; and Western Midstream.

307 *See, e.g.*, letters from CEMEX; GPA Midstream; HP; NAA; United Air; Western Midstream; and Williams Cos.; *see also* letter from Soc. Corp. Gov. (stating that "the required disclosure should be limited to climate-related risks, including value chain-related risks, reasonably likely to materially impact the registrant's financial statements and operations").

308 *See, e.g.*, letters from ABA; API; Chamber; NAM; SIFMA; and Soc. Corp. Gov.

309 *See, e.g.*, letters from Amer. for Fin. Reform, Sunrise Project *et al.*; Anthesis; CEMEX; NY City Comptroller; and TotalEnergies.

310 *See, e.g.*, letters from Amer. for Fin. Reform, Sunrise Project *et al.*; Anthesis; Bloomberg; CEMEX; Eni SpA; Hannon Armstrong (June 17, 2022); IATP; NY City Comptroller; and TotalEnergies.

311 *See, e.g.*, letters from Anthesis; Bloomberg; CEMEX; and Eni SpA; *see also* letter from Cleveland-Cliffs, Inc. (June 16, 2022) ("Cleveland-Cliffs") (opposing required disclosure of climate-related opportunities because such disclosures "are likely to be optimistic, overestimated projections at best").

risks and would provide investors with a more balanced perspective of the overall impacts of climate on a company's business and operating performance.[312]

c. Final Rules

We are adopting final rules (Item 1502(a)) to require the disclosure of any climate-related risks that have materially impacted or are reasonably likely to have a material impact on the registrant, including on its business strategy, results of operations, or financial condition, with several modifications in response to commenter concerns.[313] We disagree with those commenters who stated that a climate-related risk disclosure provision was not necessary because the Commission's general risk factors disclosure rule already requires such disclosure.[314] In our view, a separate disclosure provision specifically focused on climate-related risks will help investors better understand a registrant's assessment of whether its business is, or is reasonably likely to be, exposed to a material climate-related risk, and thereby enhance investor protection. Many commenters indicated that the Commission's current disclosure rules, including the general risk factor provision, has not provided investors with disclosure of climate-related risks and their financial impacts at the level of detail sought by investors that would make the disclosure useful for their investment or voting decisions.[315] The final rules, by contrast, are responsive to investors' need for decision-useful information regarding registrants' material climate-related risks and will help ensure investors receive more consistent, comparable, and reliable disclosures about such risks.[316]

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312 *See, e.g.*, letters from Morningstar; PwC; and World Business Council for Sustainable Development (Jun. 16, 2022) ("WBCSD").

313 *See* 17 CFR 229.1502(a).

314 *See supra* note 284 and accompanying text.

315 *See, e.g.*, *supra* note 102 and accompanying text; *infra* notes 395-397 and accompanying text.

316 *See supra* notes 244-249 and accompanying text.

Furthermore, adopting a climate-related risk disclosure rule that uses similar definitions (set forth in Item 1500) and is based on the climate-related disclosure framework of the TCFD, with which many registrants and investors are already familiar, will assist in standardizing climate-related risk disclosure and help elicit more consistent, comparable, and useful information for investors and limit the reporting burden for those registrants that are already providing some climate-related disclosure based on the TCFD framework.

At the same time, we recognize that many commenters expressed significant concerns about the scope of the proposed rules, indicating that they may elicit too much detail, may be costly or burdensome, could result in competitive harm, or may obscure other material information.[317] We have sought to address these concerns by modifying the definition of climate-related risks, by making the climate-related risk disclosure requirements less prescriptive, and by specifying the time frames during which a registrant should describe whether any such material risks are reasonably likely to manifest, as discussed below.[318]

The proposed rule would have required a registrant to describe any climate-related risks reasonably likely to have a material impact on the registrant, including on its business or consolidated financial statements. We have substituted "results of operations" and "financial condition" for "consolidated financial statements" here and in several of the final rule provisions to be more consistent with other Commission rules relevant to risk assessment, such as Item 303 of Regulation S-K regarding MD&A. We have used the term "business strategy" in the final rules to more closely align the final rules with the TCFD recommendation regarding the disclosure of the impacts of climate-related risks on strategy. These revisions do not create any

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317 *See supra* notes 283 and 285.

318 *See infra* section II.C.2.